```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

typedef struct
{
    char name[5];
    int tickets;
} Process;

int main()
{
    int n, total_tickets = 0;
    float total_T = 0.0;
    printf("1WA24CS217\n");
    printf("Enter the number of Processes: ");
    scanf("%d", &n);

    Process p[n];

    srand(time(NULL));

    for (int i = 0; i < n; i++)
    {
        printf("\nProcess %d:\n", i + 1);

        sprintf(p[i].name, "P%d", i + 1);

        printf("Tickets: ");
        scanf("%d", &p[i].tickets);
```

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    total_tickets += p[i].tickets;
    total_T += p[i].tickets;
}

printf("\n--- Proportional Share Scheduling ---\n");
printf("Enter the Time Period for scheduling: ");

int m;
scanf("%d", &m);

for (int i = 0; i < m; i++)
{
    int winning_ticket = rand() % total_tickets + 1;
    int accumulated_tickets = 0;
    int winner_index;

    for (int j = 0; j < n; j++)
    {
        accumulated_tickets += p[j].tickets;

        if (winning_ticket <= accumulated_tickets)
        {
            winner_index = j;
            break;
        }
    }

    printf("Tickets picked: %d, Winner: %s\n",
           winning_ticket,
```

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            p[winner_index].name);
    }

    for (int i = 0; i < n; i++)
    {
        printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n",
            p[i].name,
            ((p[i].tickets / total_T) * 100));
    }

    return 0;
}
```

Output:

```
1WA24CS217
Enter the number of Processes: 3

Process 1:
Tickets: 10

Process 2:
Tickets: 20

Process 3:
Tickets: 30

--- Proportional Share Scheduling ---
Enter the Time Period for scheduling: 5
Tickets picked: 10, Winner: P1
Tickets picked: 22, Winner: P2
Tickets picked: 2, Winner: P1
Tickets picked: 21, Winner: P2
Tickets picked: 58, Winner: P3

The Process: P1 gets 16.67% of Processor Time.

The Process: P2 gets 33.33% of Processor Time.

The Process: P3 gets 50.00% of Processor Time.

Process returned 0 (0x0)   execution time : 19.931 s
```